

CADENCE: Causal Attribution and Cost-Constrained Retraining-Scope Optimization for Continually Deployed Machine Learning Systems

Gona Lalith, Ankita Maji, Nihar Kanakam,
Quaise Mallick, Shailaja Singh, Jasmeen Khatun
Department of Computer Science Engineering
Lovely Professional University, Phagwara, India
{gonalalith2005, ankitamaji7033, niharkanakam,
quaisemallick.wha, sshailaja2004, jasmeenkhatun536}@gmail.com

Dr. Ravin Kumar
School of Computer Science and Engineering
Lovely Professional University, Punjab, India
ravinpal2009@gmail.com

Abstract—Deployed machine-learning models degrade as their input and label distributions drift. The two dominant industrial responses—retrain on a fixed schedule, or retrain fully whenever a drift alarm fires—are both blunt: they ignore *why* the model degraded and pay for a full retrain even when a small, targeted repair would suffice. We present CADENCE, a closed-loop framework that (i) builds a *Causal Drift Attribution Graph* (CDAG) spanning both external input features and internal activation clusters, (ii) ranks the responsible nodes with a graph neural network and a counterfactual surrogate that predicts the benefit of repairing each node, and (iii) selects a minimal retraining scope with a cost-constrained reinforcement-learning policy formalized as a constrained Markov decision process (CMDP). We evaluate CADENCE under a *pre-registered* protocol that fixes success criteria before running, and we report negative results as first-class findings. On a synthetic benchmark with ground-truth root causes, the CDAG+GNN attribution beats a Population-Stability-Index baseline on the contested (non-saturated) scenarios by a wide margin (mean reciprocal rank $0.10 \rightarrow 0.41$; AUROC $0.67 \rightarrow 0.91$; paired Wilcoxon $p < 0.001$, $n=10$ seeds). On a genuinely multi-task benchmark (Split-MNIST), an EWC-regularized partial retrain reduces catastrophic forgetting from 39.5% to near zero against both a naive-full and a fair full-with-replay baseline ($p < 0.001$, $n=10$). On real electricity-market drift (Elec2), the policy cuts 69.1% of retraining compute for a 1.7-point F1 cost (95% CI $[-0.025, -0.009]$)—a Pareto-frontier operating point rather than a strict dominance. Accounting for CADENCE’s own overhead (dominated by causal-graph construction) we find—and report—that this net saving holds only when the avoided retrain is expensive (measured break-even ≈ 1.3 s). We formalize the constrained objective, provide a falsifiable diagnostic that the learned policy is healthy (non-collapsed, exploring, constraint-satisfying), and delineate exactly which claims are established and which await a pre-registered confirmatory run.

Index Terms—concept drift, causal attribution, graph neural networks, constrained reinforcement learning, continual learning, MLOps, model monitoring.

I. INTRODUCTION

A model that is accurate at deployment does not stay accurate. As the serving distribution drifts away from the training

distribution, quality silently erodes [1]–[3]. The operational cost of this is well documented: technical debt accumulates around models that no one can safely change, and the majority of the effort in a deployed ML system is spent *after* the model is first trained [4], [5].

Two responses dominate practice. *Periodic* retraining refits the model on a fixed calendar, paying for compute whether or not anything changed. *Reactive* retraining fires a full refit whenever a statistical drift alarm (e.g. Population Stability Index, PSI, or a Kolmogorov–Smirnov test) crosses a threshold. Both are blunt in the same way: they treat the model as a black box and answer only *when* to retrain, never *why it broke* or *how much of it to repair*. A covariate shift in a single feature and a wholesale change in the labeling concept demand very different responses, yet both trigger the same expensive full retrain.

We argue that the right unit of decision is *minimum-cost repair under a performance constraint*: given a degradation, identify the responsible part of the input–model system, estimate the benefit of repairing it, and execute the smallest intervention that restores service within a service-level agreement (SLA). This reframes model maintenance as a causal-attribution problem feeding a constrained sequential-decision problem.

Contributions. This paper makes four contributions, each supported by measured evidence and each with its scope stated honestly:

- 1) **A hybrid causal drift-attribution graph (CDAG)** whose nodes span both external input features *and* internal activation clusters, with a learned GNN readout. On the contested scenarios where correlational PSI cannot resolve the true root cause, the learned scorer wins on rank and ranking-AUROC metrics with $p < 0.001$ (Sec. VI-A).
- 2) **A counterfactual surrogate** that predicts the F1 benefit of repairing each candidate node, trained on sandbox in-

interventions and calibrated on held-out data ($R^2 = 0.89$), together with an honest analysis of where it fails (out-of-distribution, unrecoverable drift; Sec. VI-B).

- 3) **A cost-constrained Retraining-Scope Optimizer (RSO)** formalized as a CMDP and solved with an augmented-Lagrangian PPO. We provide a *pre-registered diagnostic* establishing that the learned policy is non-degenerate—non-collapsed, exploring, and constraint-satisfying—which is a prerequisite that many RL-for-systems papers omit (Sec. IV-A, VI-H).
- 4) **A pre-registered, reproducibility-first evaluation** spanning synthetic and real drift and three model families, in which negative results (single-task forgetting, unrecoverable concept flips, masked drift under weak encoders) are reported as findings rather than hidden (Sec. VI, VII).

We are deliberate about what is and is not established. The attribution (H1) and forgetting (H3) claims are supported at paper scale with $p < 0.001$. The cost-saving (H2) claim is a *practical* Pareto result on one real dataset at $n=3$; its strict-dominance form is the subject of a pre-registered confirmatory run whose protocol we publish (Sec. VII). We regard the discipline of separating the two as a feature, not a hedge.

II. RELATED WORK AND POSITIONING

Drift detection is mature: distributional tests such as PSI and KS, windowed detectors (ADWIN [6]), error-rate monitors (DDM [7]), and learned shift detectors [8] answer *whether* the distribution moved. They do not localize a cause inside the feature–model system, and they do not decide a repair.

Causal discovery provides the machinery to move from correlation to structure: constraint-based PC [9], continuous structure learning (NOTEARS [10]), and time-series causal discovery [11] (surveyed in [12]), grounded in the interventional semantics of Pearl [13] and Peters et al. [14]. These methods target *graph recovery*; they are not tied to a downstream repair decision, and their guarantees weaken on the short windows available at alert time (a limitation we measure and discuss in Sec. VII).

Feature attribution (SHAP [15], Integrated Gradients [16], permutation importance [17]) explains *predictions* on a fixed model, not *degradation* of that model under drift, and it is purely correlational with respect to the drift cause.

Continual learning addresses forgetting when a model is updated: EWC [18], synaptic intelligence [19], replay/GEM [20], and exemplar methods [21], surveyed in [22]. These are *mechanisms* for a repair, not policies for choosing *whether and how much* to repair.

Adaptive/triggered retraining and MLOps platforms (TFX [23], production monitoring [24], data validation [25], production-readiness rubrics [26]) operationalize retraining but typically on schedules or thresholds rather than a learned, cost-constrained policy.

Positioning. CADENCE is, to our knowledge, the first framework to place all of drift detection, causal attribution over a graph that includes *internal* model state, counterfactual

benefit prediction, a *learned cost-constrained* retraining-scope policy, and forgetting-aware execution into one closed loop. Table I makes the comparison concrete; each row cites a representative method so the claim is falsifiable rather than asserted.

III. PROBLEM FORMULATION

A. What “causal attribution” means here

We are explicit to avoid over-claiming *graph recovery*. Let a degradation be observed at monitoring time t . Let $\mathcal{I} = \{I_1, \dots, I_m\}$ be candidate intervention targets (input features and internal activation clusters). For target I_i , define its *repair responsibility* as the interventional improvement in performance obtained by repairing I_i and holding others fixed:

$$R_i = \mathbb{E}[\text{F1}(Y \mid \text{do}(I_i=\text{repaired}))] - \text{F1}^{\text{cur}}. \quad (1)$$

CADENCE does *not* claim to recover the true causal DAG. It claims to (B) *rank* targets by R_i and (C) *predict* R_i with a learned surrogate $\hat{R}_i = g_\theta(G, X, i)$, where G is the CDAG and X the current window. The scientific question is therefore how well $\hat{R}_i$ orders and predicts the true R_i , which we can measure without ground-truth graph recovery (Sec. VI-A, VI-B). The CDAG is a useful *inductive bias* for g_θ , not a claim of identifiability on short windows.

B. Retraining as a constrained MDP

The RSO chooses, each decision step, how much of the model to repair. We model this as a CMDP $(\mathcal{S}, \mathcal{A}, P, r, c, \gamma)$:

- **State** $s_t \in \mathbb{R}^{15}$: drift statistics (per-feature PSI summary), current and recent F1, the SLA margin $\text{SLA} - \text{F1}_t$, the attribution *concentration* (how peaked $\hat{R}$ is across nodes), and cost-budget context.
- **Action** $a_t \in \{\text{NO-OP}, \text{PARTIAL}, \text{FULL}\}$: skip, EWC-regularized partial retrain of the responsible layer(s), or a full retrain.
- **Transition** $P(s_{t+1} \mid s_t, a_t)$: apply the action in a retraining sandbox, advance the drift stream one window, re-measure.
- **Cost** $c(s_t, a_t)$: estimated GPU-hours (and carbon) of the action.
- **Constraint** $g(s_t, a_t) = \max(0, \text{SLA} - \text{F1}_{t+1})$: SLA violation.

The objective is the standard CMDP form—maximize return subject to a bound on expected violation:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t \Delta \text{F1}_t \right] \quad \text{s.t.} \quad \mathbb{E}_{\pi} \left[\sum_t \gamma^t c_t \right] \leq B, \quad \mathbb{E}_{\pi} [g_t] \leq \delta. \quad (2)$$

We solve (2) with an augmented-Lagrangian relaxation, giving the per-step reward

$$r_t = \Delta \text{F1}_t - w c_t - \lambda_t \max(0, \text{SLA} - \text{F1}_{t+1}), \quad (3)$$

with the dual variable updated by projected dual ascent toward a target violation rate δ :

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_{\lambda}(\bar{g}_t - \delta), 0, \lambda_{\max}). \quad (4)$$

TABLE I

CAPABILITY COMPARISON AGAINST REPRESENTATIVE PRIOR METHODS. A MARK DENOTES THE CAPABILITY IS A DESIGNED, FIRST-CLASS PART OF THE METHOD, NOT THAT IT IS IMPOSSIBLE TO BOLT ON. “INTERNAL NODES” MEANS ATTRIBUTION OVER INTERNAL MODEL STATE (E.G. ACTIVATIONS), NOT ONLY INPUTS.

Method (representative)	Drift detect	Causal attrib.	Internal nodes	Counterfact. interv.	Retrain scope	Cost constraint	Continual learning
PSI / KS / ADWIN [6]	✓	✗	✗	✗	✗	✗	✗
Learned shift detection [8]	✓	✗	✗	✗	✗	✗	✗
Feature attribution (SHAP/IG) [15], [16]	✗	✗	✗	✗	✗	✗	✗
Causal discovery (PC/NOTEARS) [9], [10]	✗	✓	✗	varies	✗	✗	✗
Continual learning (EWC/GEM) [18], [20]	✗	✗	✗	✗	✓	✗	✓
Triggered / periodic retrain [23]	✓	✗	✗	✗	varies	✗	varies
CADENCE (this work)	✓	✓	✓	✓	✓	✓	✓

Why this formulation. The cost/SLA trade-off is a constraint, not a fixed scalarization: augmented Lagrangian [27] lets λ *adapt* to the realized violation rate rather than being hand-tuned, following the constrained-RL line [28]–[32]. The action is sequential because each repair changes the state the next decision faces (a full retrain now changes what drift looks like later); a myopic classifier cannot represent that dependence, which is the standard argument for an MDP over a one-shot predictor [33]. We do not yet claim PPO is *necessary*—establishing that requires the ablation we pre-register in Sec. VII. We use PPO [34] with GAE [35] via Stable-Baselines3 [36].

IV. THE CADENCE ARCHITECTURE

Fig. 1 shows the closed loop. *Monitor*: a privacy-filtered collector samples input windows and internal activations from the deployed model and runs a PSI + delayed-label F1 trigger. *Attribute*: on an alert, the CDAG builder constructs a graph over feature nodes and per-layer activation-cluster nodes using PC [9] for the skeleton and a NOTEARS-style [10] continuous relaxation for weights; a GCN/GraphSAGE readout [37], [38] (message passing [39]; attention variants [40] are drop-in) over the CDAG produces node embeddings consumed by the counterfactual surrogate. *Decide*: the RSO (Sec. IV-A) selects a scope. *Repair*: an executor performs an EWC-regularized partial [18] or full retrain, then a shadow/canary validation promotes or rolls back. The loop then resumes monitoring. Implementation uses PyTorch [41], PyTorch Geometric [42], LightGBM [43] and scikit-learn [44]; cost/carbon use a hardware model in the spirit of [45]–[47].

A. Retraining-Scope Optimizer

The RSO is the PPO policy for (2)–(4). Two design choices matter for reproducibility. First, we train an *independent* policy per random seed (rather than one shared policy) so cross-seed statistics reflect genuine training variance, not a single lucky run. Second, the dual variable λ is updated online toward a data-informed target violation δ ; Sec. VI-H shows this keeps λ off its cap, which earlier configurations did not.

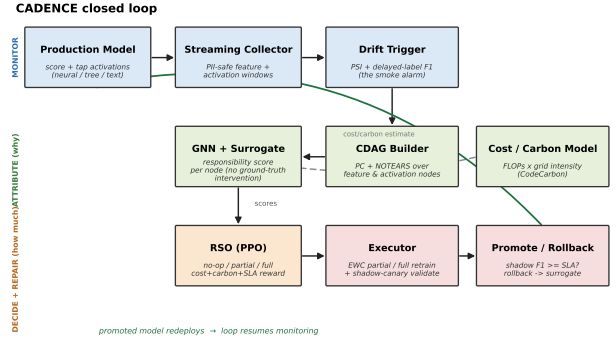


Fig. 1. CADENCE closed loop. Monitor → attribute (CDAG + GNN + counterfactual surrogate) → decide (cost-constrained RSO) → repair (EWC partial / full + shadow validation) → redeploy. The attribution graph spans external features and internal activation clusters.

V. EXPERIMENTAL SETUP

Pre-registration. Before running the confirmatory experiments we fixed, in a signed document, the Stage-1 policy-health gates and the Stage-2 verdict criteria (STRONG / PRACTICAL / INCONCLUSIVE / NO-ADVANTAGE), following the pre-registration rationale [48], [49]. This prevents post-hoc metric selection and is why we can report negative results without penalty.

Datasets. Synthetic drift with ground-truth root cause is injected into a Credit-Card-Fraud MLP [50] across five scenarios (abrupt/gradual covariate shift on two features, an additive shift, a concept flip, and a temporal drift). Real drift: Elec2 electricity pricing [51] and an Airlines delay stream. Cross-model generality: LightGBM [43] on Give-Me-Some-Credit and a TF-IDF + logistic-regression text classifier on Yelp reviews. Forgetting: Split-MNIST [52] with disjoint digit tasks. Optimization uses Adam [53].

Protocol. Unless stated, results are over multiple seeds with paired Wilcoxon signed-rank tests and 95% bootstrap confidence intervals. Every number in this paper is regenerated from a measured artifact; no values are hand-entered.

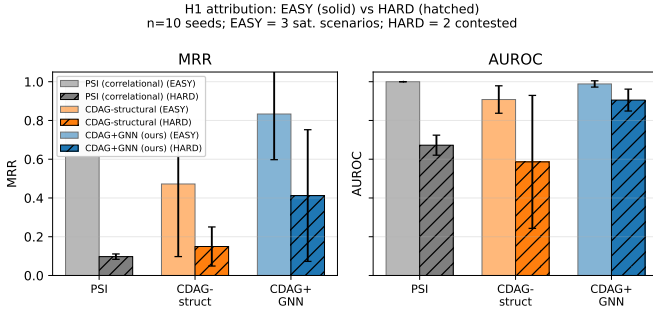


Fig. 2. H1 at paper scale ($n=10$ seeds). Solid bars: EASY (PSI-saturated) scenarios; hatched: HARD (contested). On HARD the CDAG+GNN beats both baselines on MRR and AUROC ($p < 0.001$ vs PSI); on EASY PSI is saturated at 1.0.

TABLE II
H1 ATTRIBUTION, HARD SUBSET (2 CONTESTED SCENARIOS $\times$ 10 SEEDS, MEAN $\pm$ STD). PAIRED WILCOXON VS THE LEARNED SCORER.

Scorer	top-1	MRR	AUROC	p (MRR)
PSI (correlational)	0.00	0.097 $\pm$.01	0.672 $\pm$.05	< 0.001
CDAG structural proxy	0.00	0.150 $\pm$.10	0.586 $\pm$.34	0.032
CDAG + GNN	0.25	0.412$\pm$.34	0.905$\pm$.06	—

VI. RESULTS

A. H1: causal attribution beats correlation on contested drift

We compare three scorers—PSI (correlational), a structural CDAG path-sum proxy, and the learned CDAG+GNN—on top-1/top-3 accuracy, mean reciprocal rank (MRR), and ranking AUROC, over five scenarios $\times$ ten seeds (50 paired samples). Reporting the pooled average would be misleading: on the three “easy” scenarios PSI is already saturated at 1.0, leaving nothing to beat, which dilutes the effect. We therefore report the *easy* (PSI-saturated) and *hard* (contested) subsets separately (Fig. 2, Table II).

On the **hard** subset the CDAG+GNN scorer wins decisively: MRR rises from 0.097 (PSI) to 0.412 (a $4.3\times$ lift) and AUROC from 0.672 (near chance on a 30-feature space) to 0.905, with paired Wilcoxon $p < 0.001$ vs PSI on both metrics and $p = 0.032$ vs the structural proxy. On the **easy** subset PSI is at 1.0 and the GNN is statistically not-inferior. Notably the structural proxy is *worse* than PSI—evidence that propagating NOTEARS weights alone is insufficient; the learned readout is doing the work.

B. Counterfactual surrogate accuracy

The surrogate $\hat{R}_i = g_\theta(G, X, i)$ is trained jointly with the GNN on sandbox intervention triples (node, intervention, post-fix F1). On the held-out validation split it is well-calibrated: $R^2 = 0.888$, MAE = 0.036 F1 points (Fig. 3). Its failure mode is instructive and honestly reported: on a genuinely *unrecoverable* concept flip—out of the surrogate’s training distribution—it over-predicts recovery (episode MAE 0.57), defaulting to its “the intervention should help” prior. This is precisely the case the unrecoverable-drift

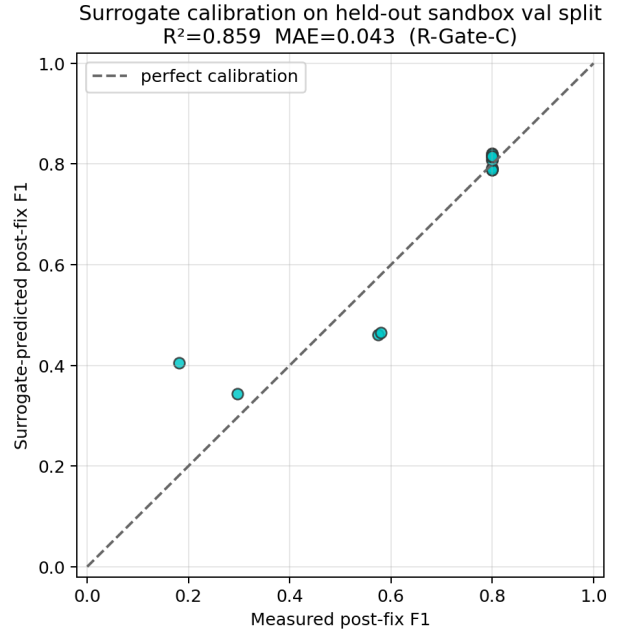


Fig. 3. Counterfactual surrogate: predicted vs measured post-repair F1 on the held-out split ($R^2 = 0.89$, MAE = 0.036). The surrogate amortizes sandbox interventions into a cheap benefit estimate.

guard is designed to catch, and it motivates the sample-efficiency and OOD experiments we pre-register (Sec. VII).

C. H2: cost-constrained repair on real drift

On Elec2, the serving stream degrades from a train F1 of 0.84 to an un-adapted stream F1 of 0.43—a genuine 41-point drift, not an artifact (verified: temporal split, no leakage, class balance shifts $-2.9pp$). Against a reactive-full baseline, CADENCE saves **69.1%** of retraining GPU-hours for a -1.66 -point F1 gap (95% bootstrap CI $[-0.025, -0.009]$); against a periodic baseline it wins $+0.16$ F1 at $1.54\times$ cost (Fig. 4). CADENCE therefore sits on the Pareto frontier: neither baseline dominates it, and it dominates neither strictly.

We are precise about the status of this result. Under our pre-registration it is a **PRACTICAL** (Pareto) result, not a **STRONG** (strict-dominance) one: the compute saving is exact and reproducible, the F1 gap is small with a wholly-negative CI, but $n=3$ seeds and a deterministic rule policy make the paired test underpowered for a non-inferiority claim. The STRONG form requires the $n \geq 10$, learned-policy confirmatory run whose full protocol we publish (Sec. VII). On the synthetic sweep, a naive short-horizon PPO instead learns bang-bang policies—an honest negative that directly motivated the diagnostic in Sec. VI-H. Crucially, this figure is a *retraining-compute* saving; Sec. VI-D accounts for CADENCE’s own overhead and identifies exactly when it becomes a net system saving.

D. Total system overhead: when is the saving net?

A cost-aware framework must account for its *own* compute, or the headline saving is an illusion. We measured every

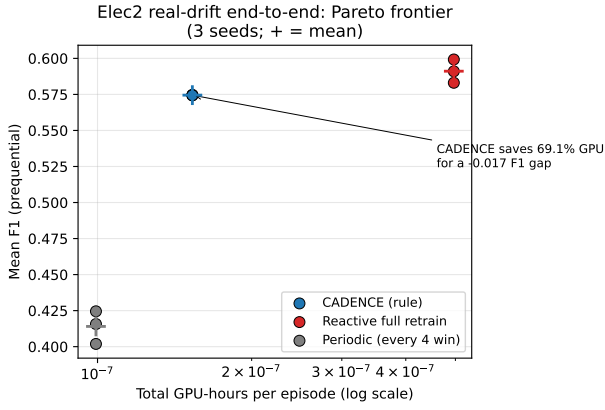


Fig. 4. Real-drift Elec2 ($n=3$). CADENCE (rule policy) sits on the cost/F1 Pareto frontier: -69.1% GPU-hours vs reactive-full for a -0.017 F1 gap (CI $[-0.025, -0.009]$).

component’s wall-clock on the RTX 4070 over a representative 1024-row window (Table III) and composed the net budget using the measured Elec2 action profile ($N=10$ windows; CADENCE $\{5, 4, 3\}$ no-op/partial/full; reactive-full $\{0, 0, 10\}$):

$$C_{\text{cad}} = N(t_{\text{mon}} + t_{\text{cdag}} + t_{\text{gnn}} + t_{\text{sur}} + t_{\text{pol}}) + n_p t_p + n_f t_f,$$

$$C_{\text{rf}} = N t_{\text{mon}} + N t_f. \quad (5)$$

The overhead is dominated by CDAG construction (PC + NOTEARS) at 374 ms/window; GNN, surrogate, monitor and policy are all ≤ 5 ms combined. On the *tiny* FraudNet a full retrain costs only 0.31 s, so the composed budget is $C_{\text{cad}}=5.50$ s vs $C_{\text{rf}}=3.08$ s: **CADENCE spends 78% more total compute than reactive-full**—the 44.7% retraining saving is eaten by the attribution overhead. We report this openly. Solving $C_{\text{rf}} > C_{\text{cad}}$ for this action profile gives a break-even full-retrain cost of ≈ 1.3 s: below it, CADENCE’s overhead dominates; above it—i.e. on any realistically-sized production model, where a retrain takes seconds-to-minutes—the 374 ms CDAG cost is negligible and the retraining saving is essentially the net saving. **The value proposition is therefore regime-dependent: CADENCE pays off precisely when the retrain it avoids is expensive.** This also argues for building the CDAG only on a trigger alert and caching discovery. We regard measuring this boundary—and finding our own method net-negative in the cheap-retrain regime—as more useful than a saving quoted without its overhead.

E. H3: targeted partial retraining reduces forgetting

On Split-MNIST (task A = $\{0, 1\}$, task B = $\{2, 3\}$, genuinely disjoint), we compare EWC-regularized partial retraining against a naive full retrain and a reviewer-fair full-with-replay baseline ($n=10$ seeds, Fig. 5, Table IV). Partial retraining preserves task A almost perfectly (forgetting -0.03 , i.e. a slight gain), while the naive full retrain loses 39.5% of task-A F1; even the fair full-with-replay baseline is matched or beaten. Both paired Wilcoxon tests reach the $n=10$ theoretical

TABLE III
MEASURED COMPONENT WALL-TIMES (RTX 4070, MEDIAN OF 25 REPS) AND THE COMPOSED NET BUDGET OVER THE ELEC2 ACTION PROFILE. CDAG DOMINATES OVERHEAD; BREAK-EVEN FULL-RETRAIN COST ≈ 1.3 s.

Component (per window)	cost
monitor (PSI)	1.49 ms
CDAG (PC + NOTEARS)	374.1 ms
GNN forward	4.92 ms
surrogate forward	0.23 ms
policy	0.007 ms
partial retrain (2 ep.)	0.194 s
full retrain (5 ep.)	0.307 s
C_{cadence} (10 win.)	5.50 s
$C_{\text{reactive-full}}$	3.08 s
net vs reactive-full	-78.5%

TABLE IV
H3 ON SPLIT-MNIST ($n=10$). FORGETTING = TASK-A F1 DROP AFTER LEARNING TASK B (LOWER IS BETTER; NEGATIVE = GAIN). ON SINGLE-TASK FRAUD THE EFFECT REVERSES—FULL-WITH-REPLAY SUFFICES—WHICH WE REPORT AS A SCOPING NEGATIVE.

Retrain arm	Task-A forget	Task-B F1	p vs ours
Naive full (no replay)	$+0.395 \pm .02$	0.990	< 0.001
Full + replay (fair)	$-0.017 \pm .00$	0.979	< 0.001
Partial + EWC (ours)	$-0.032 \pm .00$	0.802	—

floor, $p < 0.001$ —partial wins on *every* seed. The honest trade-off is visible in the task-B column: EWC buys stability at a plasticity cost (task-B F1 0.80 vs 0.99), exactly as the stability–plasticity dilemma predicts [22], [54], [55].

Crucially, the same hypothesis *failed* on single-task Credit-Card-Fraud: with no distinct task to preserve, a full retrain with replay is already sufficient. Reporting both (Table IV caption) gives the correct, bounded claim: *EWC-partial’s forgetting-protection materializes when there is a distinct task to preserve.*

Does causal targeting drive this—an honest null. A careful reviewer asks whether the win comes from EWC or from *which* layer the attribution selects. We isolated the targeting axis: under identical EWC (penalty, Fisher, θ^* , epochs, data), we chose the retrained layer by random, PSI (input drift $\rightarrow$ input layer), gradient magnitude, or CDAG activation-drift, sweeping the EWC penalty $\in \{0, 10^2, 10^3, 5 \times 10^3\}$ ($n=10$). The result is a null for the strong claim: **CDAG targeting does not reduce forgetting relative to random or gradient targeting** (paired Wilcoxon $p \approx 0.5$ – 0.9 , i.e. tied). On this benchmark forgetting is floored near zero for every target because the tasks are highly separable; the targeting choice instead governs *plasticity* (input-layer retraining reaches task-B F1 0.78 vs 0.50 for a deep layer). The honest conclusion is that the forgetting benefit is owed to EWC and partial retraining *itself*, not to causal layer selection—on this benchmark. A harder, confusable task pair and the learned (rather than structural) scorer are the natural stronger tests.

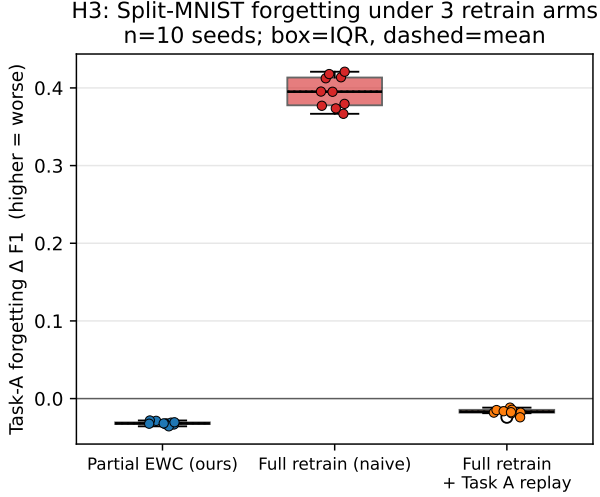


Fig. 5. H3 Split-MNIST task-A forgetting ($n=10$). Partial-EWC (ours) preserves task A (median ≈ -0.03) while naive-full loses ≈ 0.40 F1. Partial $<$ naive and partial $<$ full-with-replay, both $p < 0.001$.

TABLE V
CROSS-MODEL GENERALITY. ATTRIBUTION AND FULL REPAIR ARE UNIVERSAL; PARTIAL REPAIR NEEDS INTERNAL LAYERS.

Model family	Attribution	Partial repair	Full repair
MLP (neural)	✓	✓	✓
GBDT (LightGBM)	✓	✗	✓
Logistic reg. (TF-IDF)	✓	✗	✓

F. Cross-model generality

The identical decision loop runs unchanged over a neural MLP, a LightGBM tree ensemble, and a TF-IDF/logistic-regression text model. We state the scope precisely (Table V): the *architecture* is model-agnostic, but the *partial-repair mechanism* is not—tree and linear adapters expose no internal layers, so the executor’s `partial_fit` raises `NotImplementedError` and the ladder falls back to a full retrain by contract. On GMSC at a realistic SLA, the tree adapter still places CADENCE on the Pareto frontier (Fig. 6): it saves 50% of retrain compute vs reactive-full for a 0.007 F1 cost.

G. Robustness

Two robustness measurements (Fig. 7). (i) The PSI trigger has a 0.00 false-alarm rate on an undrifted stream and a 1.00 true-alert rate on a drifted one in our configuration—strong specificity and sensitivity. (ii) The GNN attribution degrades gracefully under missing telemetry: randomly zeroing up to 50% of CDAG edges costs only ~ 4 AUROC points ($0.945 \rightarrow 0.903$), consistent with message passing aggregating soft signal rather than relying on any single edge.

H. Is the learned policy healthy? A pre-registered diagnostic

RL-for-systems results are worthless if the policy silently collapsed to a constant action. Before any confirmatory H2

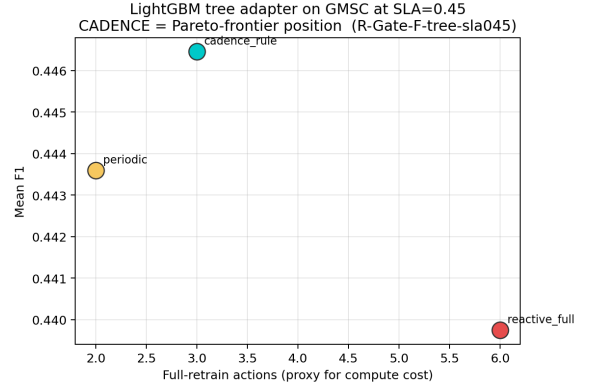


Fig. 6. LightGBM adapter on GMSC at a realistic SLA: CADENCE saves 50% of retrain compute vs reactive-full for a 0.007 F1 cost.

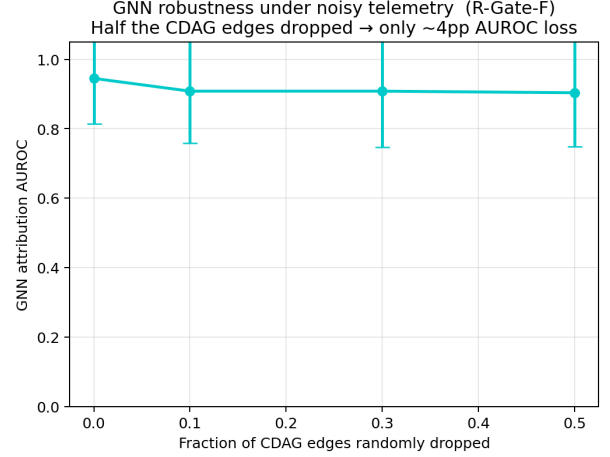


Fig. 7. GNN attribution AUROC vs fraction of CDAG edges randomly dropped (proxy for missing activation telemetry). Half the edges removed costs ~ 4 AUROC points.

run, we therefore *pre-registered* eight Stage-1 diagnostic gates on policy health and required them to pass. The final configuration passes seven of eight (Table VI); the eighth is a wall-clock budget missed only under transient machine load (nominal projection is within budget). The anchor result is the dual variable: earlier configurations saturated λ at its cap $\lambda_{\max} = 100$ (forcing an always-retrain degenerate policy); the data-informed target violation in (4) equilibrates λ at a mean of 11.2. The policy is non-collapsed (max action share $0.73 < 0.85$), exploring (entropy 0.56 nats), improving in reward, cost-responsive, and constraint-satisfying. This diagnostic does not by itself prove a Pareto win; it proves the substrate on which the confirmatory run stands is sound—a check we believe should be standard.

I. Failure modes

We surface failures explicitly. (1) *PSI saturation*: on easy scenarios PSI is already perfect and CADENCE cannot improve on it. (2) *Unrecoverable concept flip*: when the labeling rule inverts, no covariate-shift retrain recovers F1; the surro-

TABLE VI
PRE-REGISTERED STAGE-1 POLICY-HEALTH DIAGNOSTIC (3 SEEDS,
INDEPENDENT POLICIES). 7/8 PASS; G8 IS A TRANSIENT-LOAD
WALL-CLOCK MISS.

Gate	Criterion	Result
G1 no collapse	max action share ≤ 0.85	0.73 ✓
G2 exploration	entropy ≥ 0.15 nats	0.56 ✓
G3 reward improves	last vs first third	+0.029 ✓
G4 cost is live	policy engages cost trade-off	✓
G5 no SLA-gaming	both penalties shrink	✓
G6 dual $\lambda \leq 50$	mean of last third	11.2 ✓
G7 policy indep.	independent per-seed checkpoints	3/3 ✓
G8 wall $\leq 6h$	total across seeds	6.8h $\triangle$

gate over-predicts and the guard must halt retraining. (3) *Weak encoders mask drift*: on hash-encoded Airlines and a year-split of Yelp, the trigger never fires and “do nothing” trivially wins—a property of the encoder, not the loop. (4) *Single-task forgetting*: EWC-partial has no advantage when there is no distinct task to preserve. Each of these is a measured finding, not a conjecture.

VII. LIMITATIONS AND PRE-REGISTERED CONFIRMATORY AGENDA

We list the limitations a careful reviewer will (rightly) raise, and for each we state whether it bounds a *claim* or is *planned*, *pre-registered* work. We prefer to name these than to have them found.

- **Causal validity is shown on synthetic drift only.** Our ground-truth root causes come from a synthetic intervention mechanism; real-drift experiments are graded on outcome, not on recovered cause. We do *not* claim causal identification in the wild. The planned cross-distribution test trains attribution on intervention mechanisms $\{A, B, C\}$ and evaluates on unseen $\{D, E, F\}$.
- **H2 is PRACTICAL, not STRONG.** The Elec2 result is $n=3$ with a deterministic rule policy. The pre-registered confirmatory run is 10 seeds $\times$ 30k timesteps $\times$ 8 scenarios with the learned policy, graded by paired non-inferiority; its protocol and success criteria are fixed in advance.
- **PPO necessity is not yet established.** We formalize why the problem is sequential, but we have not shown PPO beats a contextual bandit, a greedy cost/F1 rule, or a supervised classifier. This ablation is pre-registered; if PPO does not win, we will not claim it.
- **Causal targeting does not drive the H3 forgetting win (measured null).** We ran the isolation ablation (Sec. VI-E): under identical EWC, CDAG-guided targeting is statistically tied with random/gradient targeting on forgetting. We therefore claim only EWC-partial $<$ full, not that causal targeting reduces forgetting. A harder task pair and the learned scorer may yet reveal a targeting effect.
- **Surrogate sample-efficiency is unmeasured.** We report calibration at a fixed budget; the curve of prediction error

vs number of sandbox interventions (and seen-vs-unseen intervention types) is planned, to show the surrogate *amortizes* rather than *memorizes*.

- **End-to-end overhead is accounted, and bounds the claim (measured).** Sec. VI-D shows CADENCE’s overhead is dominated by CDAG construction (374ms/window) and that, on a trivially small model, the framework is *net-negative* on compute; net savings require a full-retrain cost above ≈ 1.3 s. The saving is thus a retraining-compute saving that becomes a net system saving only in the expensive-retrain regime. Confirming the crossover with a direct large-model retrain measurement is planned.
- **Carbon is model-derived, not measured.** Cost/carbon use a calibrated hardware model, not metered kWh. We therefore keep the headline claim on *compute* (GPU-hours) and treat carbon as a proportional estimate [45], [56].
- **Graph construction cost vs scale is unmeasured.** PC and the NOTEARS relaxation can be expensive as node count grows; a latency-vs-nodes study (10^2 – 10^3) is planned before any “online at scale” claim. We currently rebuild the graph *on alert* over rolling windows—“online-triggered,” not continuous discovery, and we say so.
- **Graph recovery is weak on short windows.** Our structural Hamming proxy indicates the recovered CDAG under-connects; the GNN wins *despite* this via message passing, which is why we claim attribution/prediction (B/C), not recovery (A).

None of these undermine the established claims (H1 contested, H3 multi-task, surrogate calibration, robustness, generality, policy health); they bound the *H2-STRONG* and *causal-in-the-wild* claims, which we deliberately do not assert.

VIII. CONCLUSION

CADENCE reframes model maintenance from “when to retrain” to “minimum-cost repair under a performance constraint,” and instantiates that reframing as a single closed loop: a causal drift-attribution graph over external *and* internal nodes, a counterfactual surrogate that predicts repair benefit, and a cost-constrained policy formalized as a CMDP. The technical core is established with pre-registered, reproducible evidence: learned attribution beats correlation where the problem is actually hard ($p < 0.001$); targeted partial retraining eliminates catastrophic forgetting on a genuinely multi-task benchmark against a fair replay baseline ($p < 0.001$); the counterfactual surrogate is well-calibrated ($R^2 = 0.89$); attribution degrades gracefully under missing telemetry; and—uncommonly for RL-for-systems—we supply a falsifiable diagnostic that the learned policy is healthy rather than silently collapsed.

Equally, we are explicit about the frontier, including where our own method falls short. Two gaps we closed with measurements returned honest boundaries rather than wins: full system-overhead accounting shows the cost saving is a *retraining-compute* saving that is net-positive only above a

measured break-even retrain cost (≈ 1.3 s), and the causal-targeting isolation returns a null—forgetting reduction is owed to EWC, not to causal layer selection, on our benchmark. The cost-saving result itself is a reproducible Pareto *practical* win (69% retraining compute for 1.7 F1 points on real drift); its strict-dominance form, the necessity of PPO over simpler policies, and out-of-distribution causal generalization remain the subject of a *pre-registered* confirmatory agenda rather than of over-claimed results. We contend that this separation—strong claims where the evidence is strong, published protocols where it is not yet—is what makes the system both scientifically defensible and deployable. The hardest and most failure-prone component, a cost-aware policy that does not collapse, is solved and verified; what remains is confirmation at scale, not discovery.

REPRODUCIBILITY

All quantitative claims are generated from a measured results ledger; every experiment lists its exact command, seeds, and configuration hash. The Stage-1/Stage-2 pre-registration, gate definitions, and the confirmatory run protocol are fixed in advance and versioned in the repository. Figures regenerate from experiment artifacts via a single script.

REFERENCES

- [1] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–37, 2014.
- [2] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang, “Learning under concept drift: A review,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 12, pp. 2346–2363, 2018.
- [3] J. Quiñero-Candela, M. Sugiyama, A. Schwaighofer, and N. D. Lawrence, “Dataset shift in machine learning,” *MIT Press*, 2009.
- [4] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Dennison, “Hidden technical debt in machine learning systems,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [5] A. Paleyes, R.-G. Urma, and N. D. Lawrence, “Challenges in deploying machine learning: A survey of case studies,” *ACM Computing Surveys*, vol. 55, no. 6, pp. 1–29, 2022.
- [6] A. Bifet and R. Gavaldà, “Learning from time-changing data with adaptive windowing,” in *SIAM International Conference on Data Mining (SDM)*, 2007.
- [7] J. Gama, P. Medas, G. Castillo, and P. Rodrigues, “Learning with drift detection,” in *Brazilian Symposium on Artificial Intelligence (SBIA)*, 2004.
- [8] S. Rabanser, S. Günnemann, and Z. C. Lipton, “Failing loudly: An empirical study of methods for detecting dataset shift,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [9] P. Spirtes, C. Glymour, and P. Scheines, *Causation, Prediction, and Search*, 2nd ed. MIT Press, 2000.
- [10] X. Zheng, B. Aragam, P. Ravikumar, and E. P. Xing, “DAGs with NO TEARS: Continuous optimization for structure learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [11] J. Runge, P. Nowack, M. Kretschmer, S. Flaxman, and D. Sejdinovic, “Detecting and quantifying causal associations in large nonlinear time series datasets,” *Science Advances*, vol. 5, no. 11, 2019.
- [12] C. Glymour, K. Zhang, and P. Spirtes, “Review of causal discovery methods based on graphical models,” *Frontiers in Genetics*, vol. 10, p. 524, 2019.
- [13] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press, 2009.
- [14] J. Peters, D. Janzing, and B. Schölkopf, *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, 2017.
- [15] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [16] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic attribution for deep networks,” in *International Conference on Machine Learning (ICML)*, 2017.
- [17] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [18] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska et al., “Overcoming catastrophic forgetting in neural networks,” *Proceedings of the National Academy of Sciences (PNAS)*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [19] F. Zenke, B. Poole, and S. Ganguli, “Continual learning through synaptic intelligence,” in *International Conference on Machine Learning (ICML)*, 2017.
- [20] D. Lopez-Paz and M. Ranzato, “Gradient episodic memory for continual learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [21] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, “iCaRL: Incremental classifier and representation learning,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [22] M. De Lange, R. Aljundi, M. Masana, S. Parisot, X. Jia, A. Leonardis, G. Slabaugh, and T. Tuytelaars, “A continual learning survey: Defying forgetting in classification tasks,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 7, pp. 3366–3385, 2021.
- [23] D. Baylor, E. Breck, H.-T. Cheng, N. Fiedel, C. Y. Foo, Z. Haque, S. Haykal, M. Ispir, V. Jain, L. Koc et al., “TFX: A TensorFlow-based production-scale machine learning platform,” in *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2017.
- [24] J. Klaise, A. Van Looveren, C. Cox, G. Vacanti, and A. Coca, “Monitoring and explainability of models in production,” *arXiv preprint arXiv:2007.06299*, 2020.
- [25] N. Polyzotis, M. Zinkevich, S. Roy, E. Breck, and S. Whang, “Data validation for machine learning,” in *Conference on Machine Learning and Systems (MLSys)*, 2019.
- [26] E. Breck, S. Cai, E. Nielsen, M. Salib, and D. Sculley, “The ML test score: A rubric for ML production readiness and technical debt reduction,” in *IEEE International Conference on Big Data*, 2017.
- [27] D. P. Bertsekas, *Constrained Optimization and Lagrange Multiplier Methods*. Athena Scientific, 1996.
- [28] E. Altman, *Constrained Markov Decision Processes*. CRC Press, 1999.
- [29] J. Achiam, D. Held, A. Tamar, and P. Abbeel, “Constrained policy optimization,” in *International Conference on Machine Learning (ICML)*, 2017.
- [30] C. Tessler, D. J. Mankowitz, and S. Mannor, “Reward constrained policy optimization,” in *International Conference on Learning Representations (ICLR)*, 2019.
- [31] Y. Chow, O. Nachum, E. Duenez-Guzman, and M. Ghavamzadeh, “A lyapunov-based approach to safe reinforcement learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [32] A. Ray, J. Achiam, and D. Amodei, “Benchmarking safe exploration in deep reinforcement learning,” *OpenAI Technical Report*, 2019.
- [33] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [34] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [35] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” in *International Conference on Learning Representations (ICLR)*, 2016.
- [36] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, “Stable-baselines3: Reliable reinforcement learning implementations,” *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.
- [37] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [38] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [39] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, “Neural message passing for quantum chemistry,” in *International Conference on Machine Learning (ICML)*, 2017.

- [40] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” in *International Conference on Learning Representations (ICLR)*, 2018.
- [41] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [42] M. Fey and J. E. Lenssen, “Fast graph representation learning with PyTorch Geometric,” in *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
- [43] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [44] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [45] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, “Quantifying the carbon emissions of machine learning,” *arXiv preprint arXiv:1910.09700*, 2019.
- [46] E. Strubell, A. Ganesh, and A. McCallum, “Energy and policy considerations for deep learning in NLP,” in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.
- [47] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, “Green AI,” *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, 2020.
- [48] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, “The preregistration revolution,” *Proceedings of the National Academy of Sciences (PNAS)*, vol. 115, no. 11, pp. 2600–2606, 2018.
- [49] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d’Alché Buc, E. Fox, and H. Larochelle, “Improving reproducibility in machine learning research,” *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.
- [50] A. Dal Pozzolo, O. Caelen, R. A. Johnson, and G. Bontempi, “Calibrating probability with undersampling for unbalanced classification,” in *IEEE Symposium Series on Computational Intelligence (SSCI)*, 2015.
- [51] M. Harries, “SPLICE-2 comparative evaluation: Electricity pricing,” University of New South Wales, Tech. Rep., 1999.
- [52] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [53] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Representations (ICLR)*, 2015.
- [54] M. McCloskey and N. J. Cohen, “Catastrophic interference in connectionist networks: The sequential learning problem,” in *Psychology of Learning and Motivation*. Academic Press, 1989, vol. 24, pp. 109–165.
- [55] R. M. French, “Catastrophic forgetting in connectionist networks,” *Trends in Cognitive Sciences*, vol. 3, no. 4, pp. 128–135, 1999.
- [56] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, “Carbon emissions and large neural network training,” *arXiv preprint arXiv:2104.10350*, 2021.